

Generation Interconnection System Impact Study Report

For Queue Project Q20-1564

Coronado 138 kV
September 02, 2021

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1564. The Interconnection Customer proposes a 122.0 MW solar photovoltaic facility interconnecting at the Coronado 138 kV substation in the New Jersey area. The request was received on September 04, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 591 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 2 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$1,158,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
TOTALOMA JCT - VESPER 138kV	GRANDVIEW - CORONADO 138kV	100.3	212	8.8
WALNUT-ENGLE TAP - CABINET GORGE 138kV	ORANGE SUBSTATION - MOJAVE SIPHON 345kV line	100.1	464	7.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VESPER - TOALOMA JCT 138kV	Reconductor line	\$962,000	100% (\$962,000)
WALNUT-ENGLE TAP - CABINET GORGE 138kV	Reconductor line	\$1,102,000	18% (\$196,000)
	Total allocated to project		\$1,158,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.